

ADITYA PALAPATI

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Education

Indian Institute of Information Technology, Sri City, Tirupati, AP <i>B.Tech - Computer Science and Engineering</i>	08/2023 – Present CGPA: 8.05
Narayana Junior College, Vijayawada, Andhra Pradesh <i>MPC - IIT</i>	2021 – 2023 Percentage: 97.2

Technical Skills

- **Languages:** Python, JavaScript, Java, C, SQL.
- **AI/ML:** PyTorch, Hugging Face Transformers, PEFT/LoRA Fine-Tuning, Vision-Language Models, Data Cleaning & Preprocessing, Exploratory Data Analysis (EDA), Feature Extraction (TF-IDF), NLP Data Pipelines, Prompt Engineering.
- **Backend & DB:** Node.js, Express.js, RESTful API design, MongoDB (schema design, queries), PostgreSQL, MySQL.
- **Tools & Fundamentals:** Git/GitHub, Postman, Linux/Ubuntu, Shell Scripting, Tableau, Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Theory of Computation, OOP.

Projects

- Medical Image VQA using PaliGemma** | *Python, PyTorch, Hugging Face, PEFT/LoRA* 2026 – Present
- Building an end-to-end visual question answering pipeline for medical images by fine-tuning the PaliGemma vision-language model on domain-specific image-question-answer data.
 - Preprocessed and structured medical imaging datasets and corresponding QA pairs for supervised fine-tuning.
 - Fine-tuned PaliGemma using parameter-efficient techniques (PEFT/LoRA) with Hugging Face Transformers and PyTorch to adapt the model to the medical domain under limited compute.
 - Evaluating model outputs against ground-truth answers to iterate on data quality and fine-tuning configuration.
- MiniCache – In-Memory Key-Value Store (Redis Clone)** | *Node.js, TCP Sockets* 05/2026 – Present
- Built an in-memory key-value store from scratch, implementing a custom hash table, LRU eviction policy, and TTL-based key expiry via a min-heap.
 - Designed a RESP-style wire protocol over raw TCP sockets supporting core commands (GET, SET, EXPIRE, DEL); benchmarked throughput to validate design tradeoffs.
- EduAxis – School Information Management System** | *React, Node, Express, MongoDB* 01/2025 – 04/2025
- Built a full-stack RESTful application (5-person team) with role-based access control for **4 user roles** — Students, Teachers, Admin, and Super Admin.
 - Owned the teacher-student messaging and feedback module end-to-end: designed the MongoDB schema, built the Express API routes, and the corresponding React components.

Achievements

- Solved 100+ Data Structures & Algorithms problems across LeetCode and GeeksforGeeks, strengthening core problem-solving fundamentals.
- Participated in institute-level hackathons with 120+ attendees, collaborating under time constraints to build and present working prototypes.

Leadership

- Chair**, Abhisarga (Institute Fest) — IIIT Sri City 2026 – Present
- Lead planning and execution across 50+ volunteers spanning logistics, sponsorship, PR, and operations for the institute's flagship fest.
 - Spearhead public outreach and stakeholder communications, forging partnerships with external organizations, media, and sponsors to expand reach.
- Secretary**, Student Development Council (SDC) — IIIT Sri City 2025 – Present
- Coordinate events for 500+ students as cross-functional liaison across creative, admin, and external teams.
 - Manage PR across 10+ clubs and organizations, ensuring consistent communication and alignment across diverse stakeholder groups.